SYSTEM OF PARTICLES AND ROTATIONAL MOTION

1. Which of the following statements are correct about a rigid body -

(a) Ideally a rigid body has a perfectly definite and unchanging shape.

(b) Distances between different pairs of particle of rigid body do not change

(c) Inter molecular forces for a rigid body are weak

(d) No real body are truly rigid

(1) a, b, c

(2) a, b, d

(3) c,b,d

(4) All of the above are correct

2. Which of the following is correct -

(1) In pure translational motion at any instant of time every particle of the body has the same velocity

(2) In rotation of a rigid body about a fixed axis, every particle have same velocity at an instant.

(3) In rotation about a fixed axis of a rigid body different particles have different angular velocity.

(4) In rotation about a fixed axis direction of angular velocity of all particles keep on changing.

3. Which of the following is incorrect :-

(1) To determine the motion of the centre of mass of a system no knowledge of internal forces of the system is required.

(2) To determine the motion of the centre of mass of a system we need to know only the external forces on the body.

(3) The centre of mass of a system of particles moves as if all the mass of the system was concentrated at the centre of mass.

(4) The centre of mass of the fragments of the projectile changes its original trajectory after explosion

4. Select the incorrect option :-

(1) The total momentum of a system of particles is equal to the product of the total mass of the system and the velocity of its centre of mass.

(2) In pure rotation all parts of the body have the same angular velocity at any instant of time.

(3) For rotation about a fixed axis, the direction of angular velocity does not change with time if $\tau_{axis = 0}$

(4) For rotation about a fixed axis magnitude of angular velocity also remains constant

5. Which of the following is wrong :-

(1) The vanishing of the total external force and the vanishing of the total external torque are independent conditions

(2) We can have net external force without having torque

(3) We can have net torque without having net external force

(4) In a force couple acting on a body net external force is zero and also net torque is zero

6. Select the incorrect statement -

(1) The total torque on a system is independent of the origin if the total external force is zero

(2) The centre of gravity of a body coincides with its centre of mass only if the gravitational field does not vary from one part to another of the body.

(3) The angular momentum $\vec{L}$ and the angular velocity $\vec{\omega}$ should be necessarily parallel vectors.

(4) None of these

7. Which of the following is incorrect :-

(1) The centre of gravity of an external body is that point where the total gravitational torque on the body is zero.

(2) In rotational equilibrium the total external torque on the body is zero.

(3) Angular velocity $\vec{\omega}$ is a vector directed along axis of rotation.

(4) In pure translation, velocity of different parts is different.

8. In pure rolling motion on ground which of the following is wrong :-

(1) Work done by friction is zero

(2) Friction may be directed along the direction of motion

(3) Relative motion of the lowest point w.r.t. ground is zero

(4) Velocity of each point is same as that of centre of mass

9. In pure rolling of sphere on horizontal surface with constant velocity of CM :-

(1) Slipping may occur

(2) Work done by normal force is non-zero

(3) Frictional force is acting backwards

(4) Frictional force is zero

10. If a body rolls down an inclined plane without slipping -

(1) It possess only translatory kinetic energy

(2) It possess only Rotational kinetic energy

(3) No work is done by friction

(4) Gravity does no work

11. Which of the following is not an example of conservation of angular momentum :-

(1) When a person sitting on turn table suddenly folds his arms, angular velocity increases

(2) Bullet fired from gun

(3) If ice on poles of earth melts then duration of the day increases

(4) Ballet dancer streches his arms so angular velocity of rotation decreases

2. Which of the following is wrong -

(1) Radius of Gyration of a body, about an axis may be defined as the distance from the axis of a point mass whose mass is equal to the mass of the given body and moment of inertia is equal to the moment of inertia of body about same axis

(2) Moment of inertia depends on mass of a rigid body, its shape and size

(3) Moment of inertia depends on the distribution of mass about the axis

(4) Moment of inertia is independent of density of body

13. Which of the following is wrong ?

(1) A pair of equal and opposite forces with different lines of action is known as a couple

(2) Couple produces rotation without translation

(3) Moment of couple is independent of the reference point

(4) In rotational equilibrium net torque is zero only about a certain axis

14. If a particle moves in the X - Y plane, the resultant angular momentum about origin has:-

(1) Only X Component

(2) Only Y Component

(3) Both X and Y Component

(4) Only Z Component

15. Which of the following are correct :-

(a) For rotational motion, angular momentum $\vec{L}$ and angular velocity $\vec{\omega}$ need not be parallel

(b) For rotational motion, angular momentum $\vec{L}$ and angular velocity $\vec{\omega}$ are always parallel

(c) For translational motion, linear momentum $\vec{P}$ and velocity $\vec{v}$ are always parallel

(d) For translation motion, acceleration $\vec{a}$ and velocity $\vec{v}$ are always parallel.

(1) a, d (2) b, d

(3) a, c (4) c, d

16. Ideally a rigid body is a body with a perfectly definite and unchanging shape :-

(1) Distance between its two points can be changed

(2) No real body is truly rigid

(3) Rigid body has infinite inertia

(4) Rigid body can not perform translational motion

17. What is translational motion ?

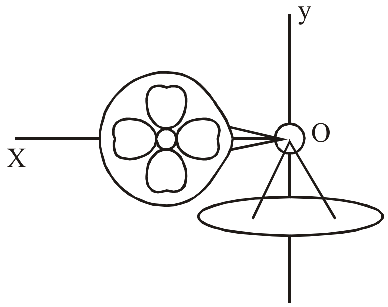
(1) Same linear velocity of all particles of body

(2) Same angular velocity

(3) both

(4) None

18. Which is axis of precession in the given figure :-

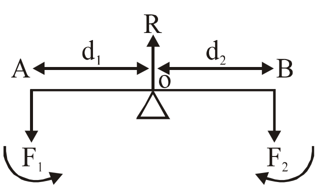


(1) OX (2) OY
(3) both OX and OY (4) None

19. In pure rotational motion :-
(1) Same linear velocity of all the particles of body
(2) Same angular velocity of all the particles of body
(3) Both (1) and (2)
(4) None of these

20. A pair of forces of equal magnitude but acting in opposite direction with different lines of action is known as -
(1) Dipole
(2) Both dipole and couple
(3) Couple
(4) None

21.



The lever force F_1 is usually some weight to be lifted. It is called the load and its distances from the fulcrum d_1 is called the load arm Force F_2 is the effort applied to lift the load; distance d_2 of the effort from fulcrum then mechanical advantage will be :-

(1) $\frac{F_2}{F_1}$ (2) $\frac{F_1}{F_2}$ (3) F_1F_2 (4) $\frac{1}{F_1F_2}$

22. What is the analogue of mass in rotational motion:-
(1) Angular momentum (2) Torque
(3) Moment of inertia (4) Inertia

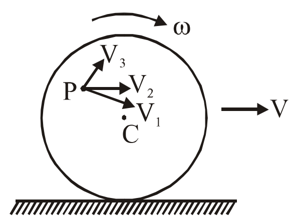
23. Part of vehicle having large moment of inertia resist the sudden increase or decrease of the speed of the vehicle known as -
(1) Engine (2) Axle
(3) Fly wheel (4) Steering wheel

24. Perpendicular axis theorem is applicable for :-
(1) One dimensional body
(2) Two dimensional body
(3) Three dimensional body
(4) One dimensional and two dimensional body

25. COM of a body (Flat body) lies at the inter section of three orthogonal axis in perpendicular axis theorem.
(1) Yes (2) No
(3) Not necessary (4) Can't Say

26. Parallel axis theorem is applicable for -
(1) One Dimensional body
(2) Two Dimensional body
(3) Three Dimensional body
(4) All type of bodies

27. Direction of resultant velocity of particle P in rolling motion :-



Pure Rolling

(1) Along V_1 (2) Along V_2
(3) Along V_3 (4) None

28. In pure rolling motion of a body if 60% of total kinetic energy is translational and 40% is rotational then the body should be -
(1) Solid sphere (2) Hollow Sphere
(3) Ring (4) Disc

29. We can apply conservation of mechanical energy in pure rolling motion of a body, even friction force is present between the surface -
(1) Mechanical energy is conserved in the presence of kinetic friction force
(2) Mechanical energy is conserved if workdone by the static friction force is zero
(3) Mechanical energy is conserved in every type of friction
(4) All are correct

30. A planet revolves around a massive star in a highly elliptical orbit. Which of the following quantity is conserved?
(1) Kinetic Energy (2) Linear momentum
(3) Angular Momentum (4) Acceleration

31. Which of the following are correct :-

(A) During rolling force of friction always act in the opposite direction of velocity of centre of mass

(B) The instantaneous speed of point of contact during pure rolling on ground is zero

(C) A wheel released on an smooth inclined plane will perform pure rolling

(D) Velocity of different points will be different in pure rolling

(1) A, B, D (2) B, C, D

(3) B, D (4) A, D

32. A particle is moving along a straight line parallel to x-axis with constant velocity. Its angular momentum about origin.

(1) Increases with time

(2) Decreases with time

(3) Is constant

(4) Only direction is constant but magnitude will change

33. If same torque was provided on a hard boiled egg & raw egg to spin on a table then :-

(1) Hard boiled egg will have higher angular acceleration.

(2) Raw egg will have higher angular acceleration

(3) Both will have same angular acceleration

(4) Information is insufficient

34. Handle of a screwdriver made wide to -

(1) Increase the weight

(2) Increase the moment of inertia

(3) Increase the torque

(4) Increase the force

35. All the points of a body performs circular motion if axis of rotation passes from :-

(1) Centre of mass of body

(2) Geometrical centre of the body

(3) Inside the body

(4) Outside the body

36. Cat always lands on its feet when dropped in any orientation due to -

(1) Increase in energy and angular momentum

(2) Decrease in energy and angular momentum

(3) By conservation of angular momentum

(4) By conservation of kinetic energy

37. A copper disc is rotating about an axis passing through centre and perpendicular to disc :-

(1) Angular velocity will increase if temperature increases

(2) Angular velocity will decrease if temperature increases

(3) Kinetic energy will increase if temperature increases

(4) Angular momentum will decrease if temperature increases

38. A body is rotating about an axis, passing through its centre of mass then its angular momentum is directed along its :-

(1) Radius (2) Tangent

(3) Circumference (4) Axis of Rotation

39. If a horizontal tube partially filled with water is rapidly rotated about a vertical axis passing through its centre, the moment of inertia of the water about axis of rotation will -

(1) Increase

(2) Decrease

(3) Constant

(4) Increase or decrease depending upon clock wise or anticlock wise sense of rotation

40. If ice of the poles will start melting the duration of day and night.

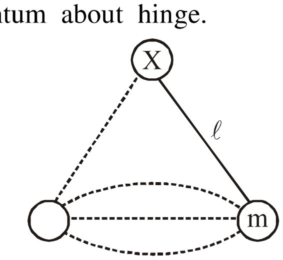
(1) Increase

(2) Decrease

(3) Not change

(4) Depend upon the mass of earth

41. A particle is tied with string and moving in a horizontal plane with constant speed. Its angular momentum about hinge.



(1) Conserved

(2) Only direction is conserved not the magnitude

(3) Only magnitude is conserved not the direction

(4) Neither magnitude nor direction is conserved

42. A rod of mass m is released on smooth horizontal surface making angle with horizontal. Then which of the following is incorrect :-

(1) Acceleration of rod along vertical is less than g.

(2) Angular acceleration of rod is not constant

(3) The angular momentum of the rod will not be constant

(4) The torque acting on the rod will be constant

43. Which of the following body has equal amount of rotational and translational kinetic energy during pure rolling?

(1) Hollow Cylinder

(2) Solid Cylinder

(3) Hollow Sphere

(4) Solid Sphere